

Matthew Oj

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TECHNICAL SKILLS

Languages: Python, SQL, JavaScript/TypeScript, PHP, HTML/CSS, C#

Frameworks & Tools: FastAPI, Next.js, React, Astro, Streamlit, Nice GUI, Laravel, Docker, GitHub Actions

Cloud & Infrastructure: AWS (Lambda, EC2, S3), GCP (Cloud Run, Vertex AI), Vercel, Netlify, CI/CD

Databases & Data Tools: PostgreSQL, SQLite, MongoDB, BigQuery, Snowflake, dbt, Apache Airflow, Snowpipe, DuckDB, XGBoost, SHAP, Vertex AI Generative Models

WORK EXPERIENCES

Senior Data Engineer (Backend Infrastructure)

Apr 2026 – present

VAFO Digital

London, UK

- Designed and led migration of backend orchestration from fragmented Snowflake SQL procedures, PHP sync services, and Lambda functions to a unified self-hosted service on AWS EC2 (Docker Compose), consolidating scheduling into a single production platform supporting 100+ services
- Led end-to-end ML pipeline builds with Data Science, integrating model training, evaluation, and registry systems into production, powering decisions across 2 brands and 60+ use cases
- Architected a distributed reporting service (Python, FastAPI, AWS Lambda, Mangum) with automated scheduled delivery across 10 client accounts, eliminating manual requests and cutting ad hoc engineering overhead
- Built a shared Python library deployed as an AWS Lambda Layer, standardising error handling and logging across 30+ production services and 4 integration targets (Klaviyo, DynamoDB, SFTP, S3), reducing duplicate code

Software Engineer, Data

Apr 2025 – Apr 2026

VAFO Digital

London, UK

- Built and hosted 3 client-facing analytics platforms (Python, Streamlit) on Snowflake, each supporting multiple apps, with high-performance querying against Snowflake event tables exceeding 200M+ rows to deliver real-time insights to non-technical stakeholders
- Developed Python AWS Lambda services syncing backend data to external systems (HubSpot, Klaviyo), supporting 250K+ users
- Contributed to a PHP/Laravel ingestion system (AWS S3) processing high-throughput REST API data, picking up the language and framework on the job to ship a production near-real-time service

Software Engineer | maisonprive.london

Apr 2026 – June 2026

Maison Privé Concierge Limited (Freelance)

Remote

- Built and deployed a luxury concierge marketing site (Astro, TypeScript) on Vercel, with SSR API routes handling enquiries across 9 services.
- Engineered a dynamic enquiry form with per-service field schemas across 3 categories, validation, rate limiting, and spam filtering, blocking 95%+ of bot submissions.
- Built a dual-sink pipeline routing enquiry to Mailgun and Airtable in real time, plus a cached editorial feed cutting load times by 80%.
- Collaborated with a second engineer in sprint cycles, leading sprint planning and Jira ticket breakdown to coordinate feature delivery

Data Engineer

Nov 2023 – Apr 2025

EssenceMediacom

London, UK

- Developed a secure web application for the Data Strategy team, implementing OAuth-based access for users across 5+ agencies
- Collaborated with third-party software engineers to build and deploy full-stack features, accelerating project delivery by 20%
- Designed and built GCP-hosted services (Python, Cloud Run) including a schema generation tool that cut manual build time by 30%, and a REST API-driven alert system detecting pipeline failures in near real-time for fast incident response

IT Support Analyst

Jan 2021 – Oct 2023

University of Greenwich

London, UK

- Developed static HTML/CSS webpages to display IT support documentation and helpline scripts, reducing resolution time for frequent issues
- Improved security compliance by 75% through multi-factor authentication resets using Microsoft Azure AD (now Microsoft Entra ID)

PROJECTS

Finance Tracker | FastAPI, dbt, PostgreSQL, Next.js, Docker | [GitHub](https://github.com)

April 2026

- Designed a production-style data stack ingesting raw financial data through dbt models into aggregated PostgreSQL marts, served via a JWT-secured FastAPI and visualised in a Next.js dashboard, all containerised with Docker Compose.
- Wrote a pytest suite covering auth, token validation, all endpoints, and pagination against a fully mocked database - 13/13 tests passing.

WPP & Google Cloud Hackathon (3rd Place) | Python, Cloud Run, Vertex AI, Vertex AI Generative Models, Streamlit

Sep 2024

- Designed and deployed a Cloud Run service pulling real-time data from public BigQuery tables to dynamically generate prompts for Vertex AI, rendering AI-generated advertising creatives through a Streamlit frontend, placed 3rd of all competing teams

Sims 4 Mod Manager | Python, SHA-256, DBPF Parsing | [GitHub](https://github.com)

April 2026

- Built a desktop mod manager scanning 88GB+ of mods with SHA-256 caching, conflict detection via DBPF package header parsing, and smart load-order assignment

EDUCATION

University of Greenwich

Sep 2020 – May 2023

Bachelor of Engineering, Civil Engineering, 2.1 Upper Second-Class Honours

London, UK

Dissertation (First Class): Assessing the vulnerability of catchments during flash flood events in South England using Machine Learning methods